LITERATURE REVIEW

The Dirichlet Portfolio Model: Uncovering the Hidden Composition of Hedge Fund Investments

Laszlo F. Korsos

PUBLICATION YEAR	2013
JOURNAL / WORKING PAPER	Working paper, arXiv preprint 1306.0938 (University of Chicago Booth School of Business)
DOI	10.48550/arXiv.1306.0938
RESEARCH AREA	Alternative Investments; hedge fund return decomposition and replication; Bayesian state space methods
SERIES	Alternative Investments
PAPER TYPE	Methodological, with an empirical application
REVIEWED BY	Arthur Faugeron
DATE REVIEWED	27 October 2026

ANALYTICAL ABSTRACT

The paper asks whether the unobservable asset class allocation of the aggregate hedge fund industry can be recovered from a monthly industry index return and the returns of eight candidate asset class indices. The Dirichlet Portfolio Model lets latent long-only weights follow a Dirichlet transition centred on buy-and-hold drift, with a Student-t observation law, filtered by sequential Monte Carlo and by a Gaussian Kalman approximation. In simulations and on the HFRI Fund Weighted Composite Index from 1995 to 2012, it tracks one-month-ahead index returns more closely than rolling constrained least squares and constrained Kalman filters, and the weights suggest rising municipal bond holdings in downturns; replicating portfolios and intraperiod risk estimates follow. The review finds the filter well built but the evidence thin: true weights are never observed, the estimated weights are exposed to return smoothing, the simulation partly favours the model by construction, and no test supports the small real-data gains.

1 Introduction

1.1 Paper identification and classification

The paper is a methodological contribution with an empirical application, written in the field of alternative investments and the subfield of hedge fund return decomposition and replication, with strong links to Bayesian time series and filtering. The version read in full is the arXiv preprint 1306.0938, accessed in its complete text form (abstract, all eight numbered sections, Tables 1 and 2, the descriptions of Figures 1 to 15, and the reference list). There is no appendix. The reviewer could read the text and tables in full but could not extract numbers from the plotted figures, so statements about Figures 8 to 14 rest on the authors' own description of them.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

It tries to recover what hedge funds hold in aggregate, month by month, when only an aggregate return series is observable. The latent object is a vector of non-negative weights on eight asset class indices that sums to one in every month.

WHY DOES THE QUESTION MATTER?

Hedge fund investors pay layered fees (the paper cites a 1 to 2 per cent management fee and a 15 to 25 per cent incentive fee, plus a median 1.5 per cent and 10 per cent at the fund of funds level), so knowing whether a cheap portfolio of liquid assets reproduces the index return is economically direct. Allocators also want intramonth risk estimates for holdings that report only monthly or quarterly.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Return based style analysis (Sharpe, 1992) and hedge fund factor models (Fung and Hsieh, 1997) already regress fund returns on asset class returns. Static or rolling versions impose constant weights and, in the unconstrained case, allow weights that are negative or larger than one. Kalman filter versions allow drift but not the budget constraint. What is missing, in the author's view, is a dynamic model that respects the simplex constraint by construction and tolerates fat tailed residuals.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

It proposes a Dirichlet transition for the weights and a Student-t observation equation. It investigates by Monte Carlo comparison with four competitors on simulated assets, on a simulated contrarian equity sector strategy built from S&P 500 sector returns, and on the HFRI index. It reports that the model has the lowest tracking error on all three, and that its weights shift towards municipal bonds and short-term Treasuries around the 2000 to 2001 and 2007 to 2012 downturns.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks whether the month by month vector of non-negative asset class weights of the value-weighted hedge fund industry can be filtered from the index return and eight palette asset returns more accurately by a Dirichlet transition, Student-t observation model than by rolling constrained least squares or constrained Kalman filters, where accuracy is measured by the error in explaining the next month index return with the forecast weights.

2.2 Hypotheses

The paper does not state numbered hypotheses; the following are the implicit ones, reconstructed by the reviewer.

- Weights of an aggregate long-only industry portfolio evolve by buy-and-hold drift plus a small compositional shock, so a Dirichlet centred on the drifted weights is the right transition law.
- Residual hedge fund returns orthogonal to the palette are fat tailed, so a Student-t observation law with six degrees of freedom outperforms a Gaussian one.
- A model that respects the simplex and learns sequentially outperforms static and Gaussian alternatives, and the advantage grows with the number of assets.
- The estimated weights carry economic information, for instance on allocation shifts during recessions, and can be used to replicate the index.

2.3 Existing literature and research gap

PAPER The author positions the work against constrained least squares (Chipman and Rao, 1964), Kalman filtering with state constraints (Chia, 1985; Simon and Chia, 2002), compositional time series (Grunwald, Raftery and Guttorp, 1993; Cargnoni, Muller and West, 1997) and particle filtering (Gordon, Salmond and Smith, 1993). The gap claimed is the absence of a dynamic, simplex valued model with non-Gaussian errors for a scalar observation.

CRITIQUE The paper does not engage with the closest hedge fund literature on dynamic exposures and replication: the nonlinear option-like payoffs of Fung and Hsieh (2001) and Agarwal and Naik (2004), the linear replication study of Hasanhodzic and Lo (2007), or Sharpe (1992) style analysis, which is the static ancestor of the whole exercise. Amenc et al. (2010) are cited only for the naive Kalman benchmark.

2.4 Claimed contribution

CRITIQUE The contribution is a coherent filtering model with a defensible transition law and a clear algorithm. It is demonstrated as a technique, but the claims that it uncovers the hidden composition of hedge fund investments and that it reveals allocation behaviour are not demonstrated, because the true weights are unobservable in the empirical application and the only validation is the fit to the same index return that drives the filter.

3 Theoretical Framework and Mechanism

3.1 State space structure and Bayesian recursion

PAPER Let $r_{HF,t}$ be the scalar index return, $r_{PA,t} \in \mathbb{R}^n$ the vector of palette asset returns and $w_t \in \Delta^{n-1}$ the latent weights at the beginning of month t. The model is

$$r_{HF,t} = f(w_t, r_{PA,t}), \qquad w_t = g(\mathcal{F}_{t-1}), \tag{1}$$

where $\mathcal{F}_t$ collects all returns and weights up to t. Because a portfolio return is the weighted sum of asset returns, the observation function is

$$r_{HF,t} = w_t' r_{PA,t} + \varepsilon_t. \tag{2}$$

Two Markov assumptions give the factorisation $p(w_0, \dots, w_t, r_{HF,1}, \dots, r_{HF,t}) = p(w_0) \prod_{\tau=1}^{t} p(r_{HF,\tau} \mid w_\tau, r_{PA,\tau})\, p(w_\tau \mid w_{\tau-1}, r_{PA,\tau-1})$. The filtering density is then the standard two step recursion

$$p(w_t \mid \mathcal{F}_t) \;\propto\; \underbrace{p(r_{HF,t} \mid w_t, r_{PA,t})}_{\text{likelihood}} \; \underbrace{\int p(w_t \mid w_{t-1}, r_{PA,t-1})\, p(w_{t-1} \mid \mathcal{F}_{t-1})\, dw_{t-1}}_{\text{prior}}. \tag{3}$$

Equation (3) says that the posterior on weights today is yesterday's posterior pushed through the transition law and then reweighted by how well each candidate weight vector explains today's index return. When the transition is linear and the noise Gaussian this is the Kalman filter; otherwise the integral is intractable, which motivates particles.

3.2 The Dirichlet transition

PAPER Without trading, a long-only portfolio drifts: asset i has value $V_{i,t} = V_{i,t-1}(1 + r_{i,t-1})$, so its weight becomes $w_{t-1,i}(1 + r_{i,t-1}) / \sum_k w_{t-1,k}(1 + r_{k,t-1})$. The paper centres a Dirichlet on this drifted vector:

$$w_t \sim \mathrm{Dir}\left(\alpha \, \frac{w_{t-1} \circ (1 + r_{PA,t-1})}{\sum_{i=1}^{n} w_{t-1,i}(1 + r_{PA,t-1,i})} \right). \tag{4}$$

w_{t-1}	Posterior (or particle) weight vector in the simplex, non-negative and summing to one.
$\circ$	Elementwise product.
$r_{PA,t-1}$	Returns of the eight palette indices in month $t-1$.
α	Scalar concentration. The Dirichlet parameter vector sums to α, since the drifted weights sum to one.
m_t	Shorthand for the drifted weight vector, the argument of $\alpha\,\cdot$ in Equation (4).

For a Dirichlet with parameter vector $\alpha = \alpha m_t$ and total $\alpha_0 = \alpha$, the standard moment formulas give

$$E[w_{t,i} \mid \mathcal{F}_{t-1}] = m_{t,i}, \qquad \operatorname{Var}(w_{t,i}) = \frac{m_{t,i}(1-m_{t,i})}{\alpha+1}, \qquad \operatorname{Cov}(w_{t,i}, w_{t,j}) = -\frac{m_{t,i}m_{t,j}}{\alpha+1}\ (i \neq j). \tag{5}$$

The first moment makes the mean a passive buy-and-hold forecast: the null of the model is that the industry does not rebalance. Active reallocation enters only through the Dirichlet noise and through the likelihood update. The second moment shows that α is a precision: the larger it is, the less the weights move per month. The paper reads α as a pseudo-count, so one new observation carries relative weight $1/(1+\alpha)$.

INFERENCE With $\alpha = 1600$ chosen in the empirical section, $1/(1+\alpha) = 0.000625$. For a weight of 12.5 per cent (one eighth, the equal weight prior) the one-month standard deviation is $\sqrt{0.125 \times 0.875/1601} = 0.83$ percentage points. If innovations were independent over 214 months, the accumulated standard deviation of a single weight would be about $0.83\sqrt{214} \approx 12$ percentage points, which is the order of the large allocation shifts the paper reports. Large shifts therefore require persistent, same direction updates from the likelihood, not isolated noise.

3.3 The Student-t observation law and the particle filter

PAPER The residual is modelled as a scale-location Student-t:

$$r_{HF,t} \sim t\big(w_t' r_{PA,t},\ \sigma_\varepsilon^2,\ \nu\big), \qquad p(r_{HF,t} \mid w_t, r_{PA,t}) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi\nu\sigma_\varepsilon^2}} \left(1 + \frac{(r_{HF,t} - w_t' r_{PA,t})^2}{\nu\sigma_\varepsilon^2}\right)^{-\frac{\nu+1}{2}}. \tag{6}$$

The sequential importance resampling algorithm initialises $w_0^{(p)} \sim \mathrm{Dir}(\alpha_0(1/n, \dots, 1/n))$ for $p = 1, \dots, P$ particles, then at each month (Step 1) propagates every particle through Equation (4) and (Step 2) resamples with probabilities

$$\omega_t^{(p)} = \frac{\left(\nu\sigma_\varepsilon^2 + \left(r_{HF,t} - w_t^{(p)\prime} r_{PA,t}\right)^2\right)^{-\frac{\nu+1}{2}}}{\sum_{\phi=1}^{P}\left(\nu\sigma_\varepsilon^2 + \left(r_{HF,t} - w_t^{(\phi)\prime} r_{PA,t}\right)^2\right)^{-\frac{\nu+1}{2}}}. \tag{7}$$

The constant of Equation (6) cancels in the ratio, which gives Equation (7). A particle that explains the index return exactly receives the largest weight; heavy tails mean that a single poorly fitted particle is down-weighted gently rather than eliminated. The paper sets $\sigma_\varepsilon^2 = 0.01$ and $\nu = 6$ and chooses $\alpha = 1600$ by the largest log marginal likelihood (Bayes factor) on a grid.

3.4 The conditionally normal approximation

PAPER To obtain a closed form, the Dirichlet is replaced at each date by a Gaussian with the same first two moments, and the observation by $r_{HF,t} \sim N(w_t' r_{PA,t}, \sigma_\varepsilon^2)$. With $\xi_{t-1} = \sum_k w_{t-1,k}(1 + r_{PA,t-1,k})$ the transition covariance is

$$\Sigma^Q_{t,ij} = \frac{w_{t-1,i}(1 + r_{t-1,i})\left(\xi_{t-1}\mathbb{1}_{i=j} - w_{t-1,j}(1 + r_{t-1,j})\right)}{\xi^2_{t-1}(\alpha + 1)}, \tag{8}$$

as the paper writes it, and the filter runs the usual prediction and update:

$$\Sigma_{t|t-1} = \Sigma_{t-1|t-1} + \Sigma^Q_t, \quad K_t = \Sigma_{t|t-1} r_{PA,t}\left(r'_{PA,t}\Sigma_{t|t-1} r_{PA,t} + \sigma^2_\varepsilon\right)^{-1}, \tag{9}$$

$$\mu_{t|t} = \mu_{t|t-1} + K_t\left(r_{HF,t} - \mu'_{t|t-1} r_{PA,t}\right), \quad \Sigma_{t|t} = \left(I - K_t r'_{PA,t}\right)\Sigma_{t|t-1}. \tag{10}$$

Equation (9) is a rank one update: a single scalar surprise corrects the whole weight vector along the direction $\Sigma_{t|t-1} r_{PA,t}$, so the corrections are largest on assets that are volatile in that month and covary with the surprise. The paper's displayed covariance contains ξ_{t-1} without the square that the Dirichlet formula implies; the reviewer treats this as a typographical slip and has not verified the code.

3.5 Competing estimators

PAPER The benchmarks are rolling-window least squares, $\hat{w}_t = (r'_T r_T)^{-1} r'_T r_{HF,T}$ over a window of k months. Imposing $\mathbb{1}'w = 1$ gives the constrained solution

$$\bar{w}_t = \hat{w}_t - (r'_T r_T)^{-1}\mathbb{1}\left(\mathbb{1}'(r'_T r_T)^{-1}\mathbb{1}\right)^{-1}(\mathbb{1}'\hat{w}_t - 1), \tag{11}$$

which still allows weights such as 120 or minus 80 under multicollinearity. Adding $w_{t,i} \geq 0$ gives the inequality constrained version solved by quadratic programming. The Kalman benchmarks use $w_t = w_{t-1} + \eta_t$ with either a covariance structure satisfying $\Sigma_0\mathbb{1} = 0$ and $Q\mathbb{1} = 0$ (CKalCov), or an unconstrained estimate projected onto the constraint surface with $\Upsilon_t = \Sigma_{t|t}\mathbb{1}(\mathbb{1}'\Sigma_{t|t}\mathbb{1})^{-1}$ and $\mu^P_{t|t} = \mu_{t|t} - \Upsilon_t(\mathbb{1}'\mu_{t|t} - 1)$ (CKalProj). Note that the random walk Kalman benchmark has no drift term, whereas the DPM centre does; part of the DPM advantage may come from this centring rather than from the Dirichlet shape.

3.6 Extensions in the paper

PAPER Short positions are admitted through a long and a short sub-portfolio, $w_t = (1 + \gamma_t)w^+_t - \gamma_t w^-_t$, with γ_t a Gaussian random walk. The paper reports $\gamma_t \approx 1$, which would mean that short sleeves add nothing. Intraperiod return and volatility, with τ units past the last report, are

$$E[r_{HF,t+\tau} \mid \mathcal{F}_{t+\tau}] = E[w_t \mid \mathcal{F}_t]' r_{PA,t+\tau}, \qquad \mathrm{Vol}[r_{HF,t+\tau} \mid \mathcal{F}_{t+\tau}] = E[w_t \mid \mathcal{F}_t]'\hat{\Sigma}_{r_{PA},t+\tau}E[w_t \mid \mathcal{F}_t] + \sigma^2_\varepsilon. \tag{12}$$

CRITIQUE The volatility expression treats weights as known at their posterior mean and omits the weight uncertainty term $\mathrm{tr}(\hat{\Sigma}\,\mathrm{Cov}(w_t))$, so it understates variance. It is also called a volatility although it is written as a variance.

3.7 Identification of the weights

INFERENCE The likelihood depends on w_t only through the scalar $w_t' r_{PA,t}$, which constrains the eight-dimensional state to a hyperplane in the simplex. One month of data cannot separate correlated assets; identification comes entirely from the dynamics, that is, from the assumption of slow movement ($\alpha = 1600$) accumulated over many months of varying return patterns. This is the structural reason that the rolling least squares estimates jump (they have no dynamics), and the reason that the Kalman estimates in the paper stay almost flat (their spherical covariance has no information about which assets move together). It also means that weight paths are partly determined by the prior, which the paper does not show with credible bands for the real data.

4 Data and Empirical Strategy

4.1 Data

PAPER The hedge fund series is the Hedge Fund Research Fund Weighted Composite Index (HFRIFWC), monthly, January 1995 to October 2012 (214 observations), net of manager fees. The palette consists of eight indices modelled on Fung and Hsieh (1997), listed in Table 1. Barclays Capital Indices supplied the fixed income index data. The simulations use six simulated assets with Student-t returns, and then daily returns of the four largest S&P 500 sectors (Technology, Financials, Health Care, Energy) from 4 January 2010 to 13 February 2013.

Table 1. **Palette asset classes used in the empirical application**

Index	Asset class	Role in the model
Barclays Municipal Bond Index	U.S. municipal bonds	Safe asset, weight rises in downturns
Barclays Short Term Treasury Bond Index	Short U.S. Treasuries	Cash proxy, large weight in all methods
Barclays Corporate High Yield Bond Index	High yield credit	Large weight until about 2000
Deutsche Bank US Dollar Long Futures Index	Dollar exposure	Currency component
Dow Jones UBS Commodity Index	Commodities	Real asset component
MSCI Emerging Markets Index	Emerging equity	Weight rises over the sample
MSCI EAFE Index	Developed ex-U.S. equity	Weight rises over the sample
MSCI US Equity Index	U.S. equity	Weight falls over the sample

Source: Table 1 of the paper (official, copied); role column summarises the authors' text.

4.2 Empirical specification and identification

PAPER The competing estimators are applied to the same data and compared through four forecast accuracy measures: forecasted RMSE (the tracking error), forecasted MAE, forecasted correlation and forecasted R-squared, all computed from $\square_t = r_{HF,t} - E[w_t \mid \mathcal{F}_{t-1}]' r_{PA,t}$. The forecast is of the weights, applied to the realised palette return of month t, so the measure is the out-of-sample fit of a replicating portfolio whose weights are set one month earlier. It is not a forecast of the index return from lagged information. Tuning parameters are $\alpha = 1600$ (selected by Bayes factor), $\sigma_\varepsilon^2 = 0.01$, $\nu = 6$ and α_0 unspecified in the text; the number of particles P is not reported.

CRITIQUE The identifying variation is time variation in palette returns; there is no exogenous variation in holdings. The estimator is a filter, not a causal design, and its validity depends on correct specification of the transition and observation laws.

4.3 Data limitations and biases

- Self-reported index: the author notes upward survivorship bias (failed funds stop reporting) and a possible downward bias from large funds that stop reporting at capacity. The net effect is unknown, and the index also contains backfill and stale pricing effects that the paper does not discuss.
- Smoothed returns: the author acknowledges that smoothing (Getmansky, Lo and Makarov, 2004) inflates the weight on short-term Treasuries; the correction is only mentioned, not reported with numbers.
- Fee asymmetry: the index is net of fees while palette returns are gross, and the budget constraint excludes an intercept, so any fee drag or alpha is forced into the weights.
- Linear payoffs: only linear exposures are modelled, although hedge funds hold option-like positions (Fung and Hsieh, 2001; Agarwal and Naik, 2004).
- Palette selection: the eight indices are chosen by the author without a selection criterion, and the text refers to investment grade corporate credit in Section 6.5 although the palette contains only high yield.
- Sample period inconsistency: the abstract states 1996 to 2012, Section 6.1 states January 1995 to October 2012.

5 Results

5.1 Simulation with simulated assets

EVIDENCE Weights are generated from the Dirichlet random walk in Equation (4) and the index is generated from the Student-t law, over 100 simulations with six assets. Median forecasted MAE is 0.0057 for the DPM, 0.0078 for CN-DPM, 0.0085 and 0.0079 for CLS and ICLS, and 0.0082 and 0.0081 for CKalCov and CKalProj, so the Kalman filters are about 43 per cent worse and CLS and ICLS 49 and 39 per cent worse. Median forecasted R-squared is 0.979 (DPM), 0.965 (CN-DPM), 0.938 and 0.944 (CLS, ICLS), and 0.960 for both Kalman filters. Increasing the number of assets raises the DPM forecast MAE at a decreasing rate, whereas CKalCov degrades almost linearly, with more extreme errors.

CRITIQUE The data generating process is the model being tested. Its advantage is therefore partly by construction, and the 0.979 R-squared is close to the ceiling set by the noise of the Student-t residual. The relevant information is the ranking of the competitors, not the size of the gap. Even so, CN-DPM (0.0078) is essentially level with ICLS (0.0079), so the Gaussian approximation of the Dirichlet gives no gain over a rolling constrained regression in this design.

5.2 Model hedge fund on S&P 500 sectors

EVIDENCE The true weights follow a contrarian rule, $w_{t,i} = \tilde{w}_{t,i} - (R_{t-1,i} - R_{t-1,m})$ with $R_{t-1,i} = \sum_{\tau=t-30}^{t-1} r_{\tau,i}$ and $R_{t-1,m}$ the cross-sectional mean, applied to four sector indices in daily data. For the Health Care sector, the DPM mean absolute error on the weight is 1.16 per cent and CN-DPM is 1.23 per cent; the OLS-type methods are 43 to 75 per cent worse and the Kalman methods 12 to 21 per cent worse. This is the only place where weights are compared with truth, and the truth here is a synthetic strategy. The reviewer notes that the rule as written does not restrict weights to be non-negative or to sum to one, so a normalisation step must exist that the text does not describe, and the number of sectors is only four.

5.3 Hedge fund index: tracking accuracy

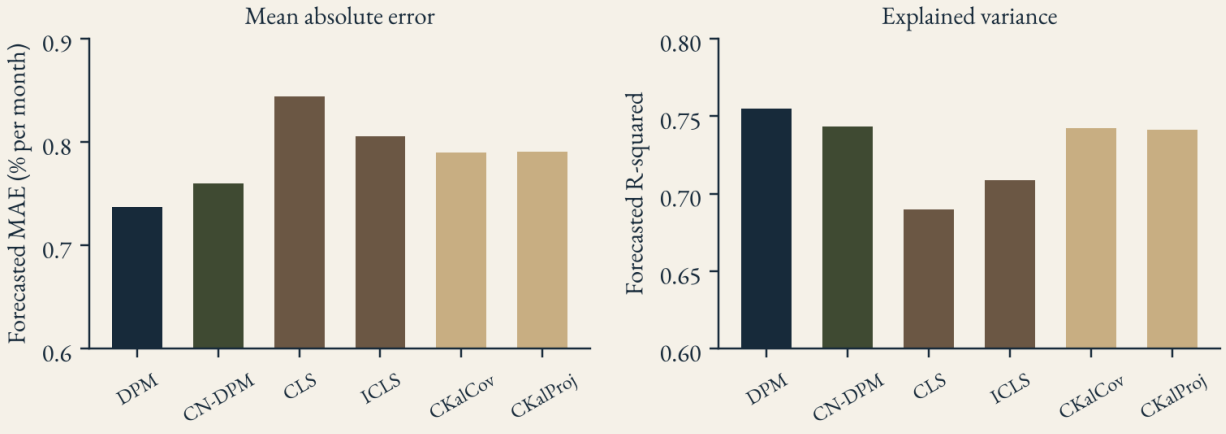
EVIDENCE Table 2 reproduces the forecasted part of the paper's Table 2. The DPM has the lowest MAE and RMSE and the highest correlation and R-squared among the six methods; CN-DPM is second on RMSE and Kalman variants are very close.

Table 2. **Replication of the HFRIFWC index with forecasted weights (monthly, January 1995 to October 2012)**

Statistic	Index	DPM	CN-DPM	CLS	ICLS	CKalCov	CKalProj
Mean	0.00748	0.00414	0.00408	0.00477	0.00447	0.00412	0.00415
Standard deviation	0.02100	0.02079	0.02084	0.02265	0.02181	0.02116	0.02125
RMSE	0	0.01049	0.01074	0.01181	0.01145	0.01076	0.01079
Mean absolute error	0	0.00737	0.00760	0.00844	0.00806	0.00790	0.00791
Correlation	1	0.88439	0.87900	0.86160	0.86529	0.87986	0.87969
R-squared	1	0.75509	0.74326	0.68966	0.70851	0.74224	0.74125

Source: Table 2 of the paper (official, copied; forecasted block). Statistics are in decimal monthly returns.

Figure 1. Forecasted mean absolute error and R-squared of the six methods on the HFRIFWC index, 1995 to 2012



Source: Table 2 of the paper (official numbers, redrawn by the reviewer). The vertical axes are truncated, so bar heights exaggerate the differences.

The numbers translate as follows. The DPM tracking error is 1.05 per cent a month, or about 3.6 per cent a year ($0.01049\sqrt{12}$, a reviewer calculation that assumes serially independent errors). Its MAE is 6.7 per cent lower than CKalCov (0.00737 against 0.00790) and 12.7 per cent lower than CLS. The mean return of the DPM replicating series is 0.414 per cent a month against 0.748 for the index, a shortfall of 0.334 percentage points a month, about 4.0 per cent a year in simple terms. Index volatility is matched almost exactly (2.08 against 2.10 per cent).

CRITIQUE The gains are modest and unaccompanied by any test. As a rough guide (reviewer estimate, assuming roughly normal errors and independence), the standard error of an MAE of 0.0074 over 214 months is about $0.0105\sqrt{1-2/\pi}/\sqrt{214} \approx 0.0004$, which is comparable with the 0.0005 gap between the DPM and CKalCov; the paired comparison would be tighter because the errors are strongly correlated across methods, but a Diebold and Mariano (1995) test or a model confidence set is needed before the ranking can be called reliable.

5.4 Estimated weights

PAPER In the DPM estimates, high yield bonds carry a sizable weight until 2000 and then investment grade credit from 2000 to 2006; weight shifts from U.S. equity to emerging and EAFE equity over the sample; short-term Treasuries and municipal bonds rise around downturns, with municipals rising in 2000 to 2001 and again from 2007 to 2012. The author reads this as a defensive reallocation consistent with lower correlation of municipal defaults with the business cycle (Appleson, Parsons and Haughwout, 2012).

CRITIQUE These results cannot be evaluated numerically because the weights appear only in figures. Short-term Treasuries and municipal bonds are the two low-volatility assets in the palette; a smoothed index with volatility of 2.1 per cent a month will load on them mechanically, which is the same artefact that the author flags for Treasuries. The municipal finding is therefore a hypothesis, not a result, until it survives a desmoothed index.

5.5 Replication and intraperiod applications

EVIDENCE The cumulative return of the replicating portfolio built from $E[w_t \mid \mathcal{F}_{t-1}]$ is plotted against the index after a fund of funds fee adjustment. The author states that DPM and CN-DPM track well, CLS does worst because of over-fitting, and the Kalman methods underperform before 2000. The author explicitly takes no stand on whether the strategy is a good investment. The intraperiod return and volatility formulas are not tested on any data at a higher frequency than monthly.

6 Robustness and Additional Tests

Table 3. **Robustness tests: concern, test, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Weights depend on the tuning parameter α	Grid on α by log marginal likelihood; variation of 70 per cent around 1600	Results not materially changed (author)	Partly. The Bayes factor holds σ_ε^2 and ν fixed, and sensitivity of the weights themselves is not shown.
Negative or explosive weights in least squares	Add non-negativity (ICLS)	ICLS is more accurate than CLS	Yes for the stated concern; it shows that constraints matter, not that dynamics matter.
Gaussian approximation	CN-DPM against DPM	Similar forecast MAE; weights more even	Partly. It shows the approximation is tolerable for tracking, but not for weight shapes.
Two Kalman constraint methods	CKalCov against CKalProj	Nearly identical	Yes, as the author proves equivalence when $Q\mathbb{1} = 0$.
Short positions	Long and short sleeves with combination factor γ_t	$\gamma_t \approx 1$, no negative exposure	Partly. No figures or numbers are given.
Smoothing of index returns	Getmansky, Lo and Makarov (2004) model	Treasury weight falls, other weights scale up proportionally	Partly. Not reported with numbers and not carried through to the municipal result.
Dependence on the simulation design	Random walk weights, then contrarian strategy on S&P sectors	DPM best in both	Partly. The first design favours the model; the second has four assets and a rule without an explicit normalisation.
Number of assets	2 to many simulated assets	DPM degrades slowly, Kalman almost linearly	Partly. Only in the DPM-generated world.

Source: reviewer's assessment of tests described in Sections 5 to 8 of the paper. Statements of result are the author's.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

Under a data generating process that matches the Dirichlet random walk, the sequential Monte Carlo filter recovers the weights and forecasts the index return better than rolling constrained least squares and the two constrained Kalman filters. On the HFRIFWC series, the DPM has the lowest forecasted RMSE and MAE of the six methods, and constraining the least squares weights to be non-negative improves them.

SUPPORTED BUT CONDITIONAL

The advantage in real data is conditional on the chosen palette, on $\alpha = 1600$, $\sigma_\varepsilon^2 = 0.01$ and $\nu = 6$, and on the use of the net-of-fee index; the MAE gap against the best Kalman filter is 6.7 per cent and untested. The claim that the advantage grows with the number of assets is conditional on the Dirichlet world.

SUGGESTIVE

That aggregate hedge fund exposure to municipal bonds rises in recessions and falls in recoveries, and that equity exposure shifts from U.S. to emerging and EAFE markets. These are patterns in weights that cannot be checked against holdings and are exposed to the smoothing artefact.

NOT ESTABLISHED

That the estimated weights are close to true industry holdings, that the replicating portfolio is a cheaper substitute for hedge fund investment (the mean shortfall is about 4 per cent a year before costs and the author takes no stand), and that the intraperiod volatility estimates are accurate (they are not tested).

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- The simplex and non-negativity constraints are built into the transition law rather than imposed afterwards, so every particle is an admissible portfolio.
- Student-t errors allow for fat tails in the unexplained return, and the particle filter allows non-Gaussian inference without linearisation.
- Both an exact (numerical) and an analytical (Gaussian) solution are provided, with the benchmarks improved by the authors' own constraints so that the comparison is not against straw men.

STRENGTHS IDENTIFIED INDEPENDENTLY

- The mean of the transition is exactly the passive drift of a long-only book. That gives the null hypothesis a clear economic content and makes α interpretable as inertia.
- The forecast measures use weights known one month earlier applied to the realised return, which is the right object for an investable replication test rather than an in-sample fit.
- The paper separates a simulation, a semi-realistic strategy and real data, which is the correct ladder of evidence, even if each rung is thin.
- The approach is general: any palette and any aggregate return series can be used, which could extend to mutual funds with disclosed holdings, where truth is available.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The accuracy of the weights is never tested against true weights on real data, and the surrogate used, the fit to the index return, is the quantity the filter is built to fit. **Why it matters.** Because the likelihood constrains only a single linear combination of eight weights, many weight paths yield the same return fit, so a good tracking error does not validate the allocation story told in Section 6. **Potential bias.** Weights can be sensitive to the prior (α, the equal weight start, the palette) and to artefacts such as smoothing, with no way to separate these from allocation. **How it could be tested.** Apply the filter to funds with disclosed holdings (for example registered liquid alternative funds or institutional portfolios with known weights), compare filtered and true weights, and report credible intervals and coverage.

OTHER WEAKNESSES

- Circular simulation: Problem, the first experiment simulates from the model. Why it matters, the ranking is a property of correct specification. Bias, favourable to the DPM. Test, generate weights from rebalancing rules unrelated to a Dirichlet (threshold rebalancing, momentum tilts, crisis de-risking).
- No statistical inference on accuracy differences: the paper reports point statistics for 214 months without standard errors or tests. A Diebold and Mariano test on the squared or absolute errors with a block bootstrap would address this.
- Linear, long-only, unlevered palette: hedge funds use leverage and options. The budget constraint forces the index return to equal a fully invested weighted average, which cannot reproduce a positive alpha or a leveraged beta; the 4 per cent annual mean shortfall may reflect this. Test by adding a free intercept and a leverage state.
- Hyperparameters set by hand: $\sigma_\varepsilon^2 = 0.01$ is a variance of 0.01, a scale of 0.10 or 10 per cent a month, which is nearly five times the monthly index volatility. See Section 8: if this is the meaning, the likelihood is nearly flat. The paper does not report the particle count or the effective sample size.

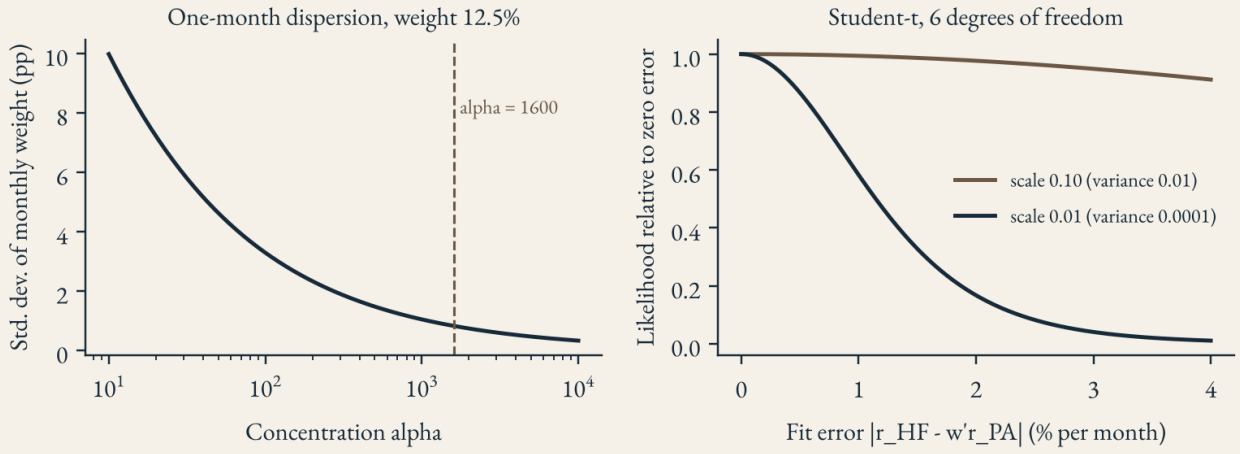
- Smoothing and illiquidity are mentioned but not integrated; the HFRI index is a fund-weighted self reported average, whereas the model is built as a value-weighted portfolio.
- Documentation: the sample start (1995 or 1996), the investment grade credit reference and the squared ξ_{t-1} in the covariance are inconsistencies; the volatility formula omits weight uncertainty.

8 Contradictions and Edge Cases

CRITIQUE Illustrative calculations below are the reviewer's, using the paper's stated parameters ($\nu = 6$, $\alpha = 1600$).

INFERENCE First, how informative is the likelihood? The importance weight of a particle with fit error e, relative to a perfect fit, is $(1+e^2/(\nu\sigma_\varepsilon^2))^{-(\nu+1)/2}$. If $\sigma_\varepsilon^2 = 0.01$ is the variance, so the scale is 0.10, a particle that misses the index by 2 per cent a month receives 0.977 of the weight of a perfect one (and 0.949 for a 3 per cent miss), so resampling hardly discriminates between particles: the posterior stays close to the Dirichlet prior, whose mean is buy-and-hold drift. If instead 0.01 is the scale itself (variance 10^{-4}), the same 2 per cent miss receives 0.167 and a 3 per cent miss 0.040, a discriminating likelihood. The first reading is consistent with the notation $t(\mu, \sigma_\varepsilon^2, \nu)$, the second with the reported tracking errors (about 1 per cent). The paper's results can be read as a successful filter only under the second reading, so the parameter definition needs clarification and sensitivity analysis. The figure below shows both curves.

Figure 2. **Dispersion of the Dirichlet transition (left) and relative likelihood of a particle by fit error (right)**



Source: generated by the reviewer from Equations (5) and (6) with the paper's parameters (alpha 1600, six degrees of freedom); not taken from the paper.

INFERENCE Second, extreme values of α. As $\alpha \to \infty$ the weights follow pure buy-and-hold drift and the data never matter. As $\alpha \to 0$ the Dirichlet puts mass on the vertices of the simplex and the filter chases the most recent return (a single asset explains it). The selected value of 1600 is far on the buy-and-hold side. A monthly weight standard deviation of 0.83 percentage points means that moving 20 points of weight into municipal bonds requires many consecutive months of consistent evidence, so quick shifts in the figures reflect sustained drift of the particle cloud rather than responsive rebalancing.

INFERENCE Third, collinearity and leverage. When the safe assets (Treasuries and municipals) are positively correlated and dominated in volatility terms by equity, a smoothed index with 2.1 per cent volatility, against about 4 to 5 per cent for U.S. equity in these decades, can be matched by a larger weight in the safe assets plus a smaller equity weight. A manager with 40 per cent equity beta at 1.5 times gross leverage and a manager with 60 per cent equity unlevered are indistinguishable once weights are forced to sum to one. This is not an error in the filter but a limit on what the weights mean.

INFERENCE Fourth, the municipal finding could come from another mechanism. During the 2008 to 2009 crisis municipal bond prices fell with illiquidity (the paper itself cites Deng and McCann, 2012), so a rising weight to municipals in 2007 to 2012 reflects a low-volatility asset with low correlation to equity, which absorbs the weight the filter must place somewhere when the index returns are smooth. A placebo test would use a synthetic index built from known weights, smoothed with an MA(2) as in Getmansky, Lo and Makarov (2004), and ask whether the filter manufactures municipal weight.

9 Further Literature

Table 4. **Literature connections**

Paper	Shared idea	Relationship	Why it matters
Sharpe (1992)	Constrained regression of fund returns on asset class indices	Preceded; the DPM is its dynamic Bayesian successor	Gives the static non-negative, sum to one estimator to which the ICLS benchmark corresponds.
Fung and Hsieh (1997)	Style factors for hedge funds	Preceded and supplied the palette logic	Shows hedge fund returns have low correlation with equity style mimics, relevant to the 75 per cent explained variance.
Fung and Hsieh (2001); Agarwal and Naik (2004)	Option-like payoffs in hedge fund returns	Contradicts the linear observation equation	Nonlinear exposure cannot be captured by long-only linear weights.
Getmansky, Lo and Makarov (2004)	Return smoothing in hedge funds	Identifies a bias in the application	Smoothing inflates low volatility weights; the author mentions it without integrating it.
Hasanhodzic and Lo (2007)	Clone hedge fund returns with liquid factors	Related replication study, linear case	Reports the cost of replication in tracking and mean, comparable to the 4 per cent shortfall here.
Amenc et al. (2010)	Bayesian state space replication	Source of the naive Kalman benchmark	The DPM is positioned as an improvement on this class.
Gordon, Salmond and Smith (1993); Doucet, Godsill and Andrieu (2000)	Sequential Monte Carlo	Methodological foundation	Support the SIR algorithm; auxiliary filters (Pitt and Shephard, 1999) could reduce Monte Carlo noise.

Paper	Shared idea	Relationship	Why it matters
Cargnoni, Muller and West (1997); Aitchison (1982)	Compositional data models	Related methodology	Handle simplex valued observations; the DPM instead has a scalar observation of a compositional state.
Carvalho et al. (2010)	Particle learning of parameters	Proposed extension	Would allow α, σ_ε and ν to be learned rather than fixed.
Fung and Hsieh (2000, 2004)	Biases in hedge fund databases and risk based benchmarks	Provide the standard treatment of the data issues	Quantify survivorship and backfill biases, absent from the paper.

Source: reviewer's reading of the bibliography of the paper and of the field (generated, not official); connections are limited to papers with a specific shared idea.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** No validation of weights against true holdings. *Why it matters:* The central claim is about hidden composition. *Existing limitation:* The surrogate accuracy measure is the return fit. *Way forward:* Test on portfolios with disclosed holdings and report credible interval coverage.
- **Gap:** No sampling uncertainty for the accuracy rankings. *Why it matters:* The real-data gains are small. *Existing limitation:* Point statistics over 214 months. *Way forward:* Diebold and Mariano tests, block bootstrap, model confidence sets.
- **Gap:** Tuning parameters are fixed. *Why it matters:* The definition of σ_ε^2 decides how informative the data are. *Existing limitation:* Hand chosen and the particle count is missing. *Way forward:* Particle learning, and sensitivity surfaces for the weights.

10.2 Literature-level gaps

- **Gap:** Joint treatment of smoothing, nonlinearity and compositional dynamics. *Why it matters:* Each distorts the weights separately. *Existing limitation:* Models treat one at a time. *Way forward:* A latent true return layer with an MA smoothing kernel and option-like factors in the palette.
- **Gap:** Net-of-cost investable replication evidence. *Why it matters:* The economic case for clones rests on net returns. *Existing limitation:* Results are gross, and mean shortfalls are large. *Way forward:* Real-time implementation with transaction costs and fees.

10.3 Methodological gaps

- **Gap:** Identification of a vector from a scalar observation. *Why it matters:* Weights are prior dependent. *Existing limitation:* Single index series. *Way forward:* Use sub-strategy indices jointly (multivariate observation), or fund level panels with hierarchical shrinkage.
- **Gap:** Time-varying concentration. *Why it matters:* Rebalancing intensity should rise in crises. *Existing limitation:* Constant α. *Way forward:* Stochastic α_t with regime switching.

11 Research Opportunities

Idea 1. Ground truth validation with disclosed holdings

RESEARCH QUESTION. Does the Dirichlet filter recover true asset class weights when applied to portfolios whose holdings are observed, and how large are the errors relative to rolling style analysis?

MOTIVATION. The main untested claim of the paper is accuracy of the weights.

HYPOTHESIS. Filtered weights lie within 90 per cent credible bands in about 90 per cent of months for slow-moving funds and fail for tactical funds.

ECONOMIC MECHANISM. Funds with frequent tactical shifts violate the slow Dirichlet drift, which makes coverage depend on turnover.

LITERATURE CONNECTION. Sharpe (1992) style analysis validation; Amenc et al. (2010).

DATA. Holdings and returns of registered multi-asset and liquid alternative funds (for example CRSP mutual fund holdings, Morningstar) and public palette indices.

VARIABLES. Monthly fund return, true asset class weights, palette returns, turnover.

METHOD. Run DPM, CN-DPM, ICLS and CKalCov on returns only; compare with true weights using mean absolute error, coverage and rank correlation.

IDENTIFICATION STRATEGY. Holdings provide the true state, so error is observed directly; turnover is used to explain heterogeneity.

EXPECTED CONTRIBUTION. First external validation of the filter, and an empirical mapping between turnover and filter error.

MAIN RISK. Registered funds are more liquid and slower than hedge funds, so the validation may not transfer.

Idea 2. Joint desmoothing and weight filtering

RESEARCH QUESTION. Does the rising municipal bond weight survive when the observation equation includes a smoothing kernel for reported returns?

MOTIVATION. The paper flags smoothing but leaves the central finding uncorrected.

HYPOTHESIS. After joint estimation the municipal and Treasury weights fall and equity weights rise.

ECONOMIC MECHANISM. Smoothed reported returns understate volatility; the budget constraint compensates through low volatility assets.

LITERATURE CONNECTION. Getmansky, Lo and Makarov (2004); Asness, Krail and Liew (2001).

DATA. HFRI composite and strategy indices, Barclays municipal and Treasury indices, equity indices.

VARIABLES. Reported index return, latent true return, smoothing coefficients, weights.

METHOD. Rao-Blackwellised particle filter with an MA(2) layer for the reported return, estimated by maximum marginal likelihood.

IDENTIFICATION STRATEGY. Serial correlation of the index identifies smoothing, and a placebo with synthetic smoothed indices tests spurious weights.

EXPECTED CONTRIBUTION. A corrected description of hedge fund allocation behaviour in recessions.

MAIN RISK. Smoothing and slow weight drift are hard to separate with one series.

Idea 3. Option-like and trend-following palette

RESEARCH QUESTION. How much of the unexplained variance of the DPM (25 per cent) is captured by adding trend-following lookback straddle returns and a long-short equity factor?

MOTIVATION. Linear long-only weights omit the main nonlinear hedge fund exposures.

HYPOTHESIS. Adding nonlinear factors raises forecasted R-squared by several points and reduces the mean shortfall.

ECONOMIC MECHANISM. Dynamic trading strategies create convex payoffs that cannot be matched by a static mix of indices.

LITERATURE CONNECTION. Fung and Hsieh (2001, 2004); Agarwal and Naik (2004).

DATA. Published trend-following factor returns, HFRI strategy indices, palette indices.

VARIABLES. Index return, nonlinear factor returns, Dirichlet weights, free intercept.

METHOD. Extend the palette and let the budget sum to a state-dependent gross exposure.

IDENTIFICATION STRATEGY. Out-of-sample forecast comparison with and without nonlinear factors for each strategy index.

EXPECTED CONTRIBUTION. A tractable replication filter for nonlinear hedge fund styles.

MAIN RISK. Gains may arise from in-sample selection of factors; strict out-of-sample design is needed.

Idea 4. Regime dependent concentration

RESEARCH QUESTION. Does the rebalancing intensity of the industry, measured by a time varying concentration α_t, fall in crises?

MOTIVATION. A constant $\alpha = 1600$ treats calm and crisis months alike.

HYPOTHESIS. α_t is lower (faster rebalancing) in months with high VIX and funding stress.

ECONOMIC MECHANISM. Deleveraging and flight to quality increase turnover during stress.

LITERATURE CONNECTION. Carvalho et al. (2010); Khandani and Lo (2007).

DATA. HFRI index, palette, VIX, TED spread, hedge fund flows.

VARIABLES. $\log \alpha_t$, volatility, spreads, flows.

METHOD. Particle learning with a stochastic volatility style law for $\log \alpha_t$ and regression on stress indicators.

IDENTIFICATION STRATEGY. Event windows around 1998, 2008 and March 2020 against calm periods.

EXPECTED CONTRIBUTION. A model-based turnover indicator derived from returns only.

MAIN RISK. Weak identification of α_t from a scalar series.

Idea 5. Net investable replication test

RESEARCH QUESTION. After transaction costs and fees, does a real-time DPM clone earn a positive net alpha relative to a fund of funds investment?

MOTIVATION. The paper leaves open whether the replication is a good investment.

HYPOTHESIS. The clone matches 80 per cent of the beta but earns a lower return than the index gross; net of fund of funds fees the difference is not significantly different from zero.

ECONOMIC MECHANISM. Hedge fund fees compensate partly illiquid and nonlinear exposures that a clone does not hold.

LITERATURE CONNECTION. Hasanhodzic and Lo (2007); Fung and Hsieh (2004).

DATA. HFRI composite, ETF or index futures returns for the palette, estimated costs.

VARIABLES. Clone and index returns, turnover, costs, fees.

METHOD. Expanding window real-time implementation; Sharpe ratio difference test with bootstrap; Diebold and Mariano on errors.

IDENTIFICATION STRATEGY. Strict use of information available at the rebalancing date; no parameter re-tuning after the fact.

EXPECTED CONTRIBUTION. Out-of-sample economic evidence for or against low cost replication.

MAIN RISK. Index biases (survivorship, backfill) distort the benchmark.

11.1 Ranking

Table 5. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Validation with holdings	4	4	4	4	5	5	26
2. Joint desmoothing	4	3	4	4	4	4	23
3. Nonlinear palette	3	4	4	4	4	4	23
4. Regime dependent concentration	5	3	4	3	3	4	22
5. Net investable replication	3	4	4	4	4	4	23

Source: reviewer's subjective scoring.

The top three are Idea 1, which removes the largest weakness of the paper with observable truth; and Ideas 2 and 5 (tied with Idea 3 at 23), chosen because the first changes the headline economic finding and the second answers the investment question the author leaves open. Idea 3 is a close alternative that is simpler to implement.

12 Learning Framework

12.1 Prerequisite concepts

- Dirichlet and Beta distributions, their moments and the pseudo-count interpretation.
- State space models, the Kalman filter and Bayes recursion.
- Particle filters: importance sampling, resampling and degeneracy.
- Constrained least squares and return based style analysis.
- Hedge fund index construction, smoothing and survivorship biases.

12.2 Core concepts learned

- A buy-and-hold drift is the natural centre of a portfolio weight process, and a single concentration parameter turns it into a probabilistic model.
- Constraints in a state space model can be built into the transition law, which avoids ex post projection.
- A scalar observation of a vector state is identified through dynamics, so weights are partly prior dependent.

12.3 Important equations

Equation (4) defines the transition law and Equation (5) its precision. Equation (7) gives the particle weights and shows that σ_ε^2 controls how much the data move the posterior. Equation (9) is the rank one Kalman correction of the Gaussian approximation.

12.4 Methods learned

Sequential importance resampling on the simplex; moment matching of a Dirichlet by a Gaussian at each date; constrained Kalman filtering by covariance structure and by state projection; Bayes factor selection of a hyperparameter; forecast based accuracy measures for replication.

12.5 Questions to test understanding

COMPREHENSION

1. Why does the mean of the Dirichlet transition equal the buy-and-hold drifted weight vector?
2. What does $\alpha = 1600$ imply for the monthly standard deviation of a 12.5 per cent weight?
3. How are the forecast accuracy measures built, and why are they not forecasts of the index return in the usual sense?

ANALYSIS

1. Why do constrained Kalman filters with a spherical covariance produce nearly flat weights?
2. How does the meaning of σ_ε^2 change the information content of the likelihood, and how would you detect the issue from the particle effective sample size?
3. Why can a smoothed index inflate weights on low volatility assets?

RESEARCH

1. How would you design a simulation that does not favour the DPM by construction?
2. What additional observations would make the weight vector identified rather than prior dependent?
3. How would you extend the model to leverage and options while keeping a simplex state?

12.6 Papers to read next

Sharpe (1992); Hasanhodzic and Lo (2007); Getmansky, Lo and Makarov (2004); Fung and Hsieh (2004); Carvalho et al. (2010).

13 Conclusion

13.1 One-sentence thesis

The paper proposes a simplex valued Bayesian filter, a Dirichlet transition around buy-and-hold drift with Student-t observation noise, to infer the hidden asset class weights of the hedge fund industry and replicate its index, and shows that it tracks the index better than constrained regressions and Kalman filters.

13.2 Three most important insights

- Building the budget and non-negativity constraints into the transition law is both conceptually cleaner and better behaved than imposing them after estimation.
- The concentration parameter α is the economic heart of the model: it sets how fast the industry is allowed to rebalance and how much the data may move the weights.
- Tracking accuracy does not validate the weights: the information in the data is a single linear combination of them each month.

13.3 Strongest evidence

The comparison on the contrarian sector strategy, where true weights are known and the DPM has the lowest weight error (1.16 per cent), together with the lowest forecasted MAE (0.00737) on the HFRI index.

13.4 Biggest weakness

No validation of the weights against true holdings, with a simulation that partly follows the model itself.

13.5 Most important research gap

Whether filtered weights are accurate when truth is observed, and how much is prior, smoothing or linearity.

13.6 Best research opportunity

Ground truth validation on funds with disclosed holdings (Idea 1), with later extension to joint desmoothing.

13.7 Three papers to read next

Sharpe (1992); Hasanhodzic and Lo (2007); Getmansky, Lo and Makarov (2004).

14 References

Agarwal, V., and Naik, N. Y. (2004). Risks and portfolio decisions involving hedge funds. *Review of Financial Studies*, 17(1), 63 to 98.

Aitchison, J. (1982). The statistical analysis of compositional data. *Journal of the Royal Statistical Society, Series B*, 44(2), 139 to 177.

Amenc, N., Martellini, L., Meyfredi, J., and Ziemann, V. (2010). Passive hedge fund replication, beyond the linear case. *European Financial Management*, 16(2), 191 to 210.

Appleson, J., Parsons, E., and Haughwout, A. (2012). The untold story of municipal bond defaults. *Liberty Street Economics*, Federal Reserve Bank of New York, 15 August 2012.

Asness, C., Krail, R., and Liew, J. (2001). Do hedge funds hedge? *Journal of Portfolio Management*, 28(1), 6 to 19.

Cargnoni, C., Muller, P., and West, M. (1997). Bayesian forecasting of multinomial time series through conditionally Gaussian dynamic models. *Journal of the American Statistical Association*, 92(438), 640 to 647.

Carvalho, C., Johannes, M., Lopes, H., and Polson, N. G. (2010). Particle learning and smoothing. *Statistical Science*, 25, 88 to 106.

Chipman, J. S., and Rao, M. M. (1964). The treatment of linear restrictions in regression analysis. *Econometrica*, 32, 198 to 209.

Deng, G., and McCann, C. (2012). Leveraged municipal bond arbitrage: What went wrong? *Journal of Alternative Investments*, 14(4), 69 to 78.

Diebold, F. X., and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business and Economic Statistics*, 13(3), 253 to 263.

Doucet, A., Godsill, S., and Andrieu, C. (2000). On sequential Monte Carlo sampling methods for Bayesian filtering. *Statistics and Computing*, 10, 197 to 208.

Fung, W., and Hsieh, D. A. (1997). Empirical characteristics of dynamic trading strategies: The case of hedge funds. *Review of Financial Studies*, 10(2), 275 to 302.

Fung, W., and Hsieh, D. A. (2000). Performance characteristics of hedge funds and commodity funds: Natural versus spurious biases. *Journal of Financial and Quantitative Analysis*, 35(3), 291 to 307.

Fung, W., and Hsieh, D. A. (2001). The risk in hedge fund strategies: Theory and evidence from trend followers. *Review of Financial Studies*, 14(2), 313 to 341.

Fung, W., and Hsieh, D. A. (2004). Hedge fund benchmarks: A risk based approach. *Financial Analysts Journal*, 60(5), 65 to 80.

Getmansky, M., Lo, A. W., and Makarov, I. (2004). An econometric model of serial correlation and illiquidity in hedge fund returns. *Journal of Financial Economics*, 74(3), 529 to 609.

Gordon, N., Salmond, D., and Smith, A. F. M. (1993). Novel approach to nonlinear and non-Gaussian Bayesian state estimation. *IEE Proceedings F*, 140, 107 to 113.

Grunwald, G. K., Raftery, A. E., and Guttorp, P. (1993). Time series of continuous proportions. *Journal of the Royal Statistical Society, Series B*, 55(1), 103 to 116.

Hasanhodzic, J., and Lo, A. W. (2007). Can hedge fund returns be replicated? The linear case. *Journal of Investment Management*, 5(2), 5 to 45.

Kalman, R. E. (1960). A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1), 35 to 45.

Khandani, A. E., and Lo, A. W. (2007). What happened to the quants in August 2007? *Journal of Investment Management*, 5, 5 to 54.

Korsos, L. F. (2013). The Dirichlet portfolio model: Uncovering the hidden composition of hedge fund investments. arXiv preprint 1306.0938. Version read: arXiv full text as accessed on 27 October 2026.

Pitt, M. K., and Shephard, N. (1999). Filtering via simulation: Auxiliary particle filters. *Journal of the American Statistical Association*, 94, 590 to 599.

Sharpe, W. F. (1992). Asset allocation: Management style and performance measurement. *Journal of Portfolio Management*, 18(2), 7 to 19.

Simon, D., and Chia, T. L. (2002). Kalman filtering with state equality constraints. *IEEE Transactions on Aerospace and Electronic Systems*, 38(1), 128 to 136.